SN 2024pba as a Two-Event Interacting Supernova: Model Scenario and Current Evidence

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Abstract

SN 2024pba is modelled as *two interacting explosion events* rather than a single supernova: an earlier Type IIb explosion (SN_A) that left fast, metal-rich ejecta, and a later event (SN_B) whose dense shell is launched into the space inside the aged SN_A ejecta and drives an embedded shock. This note summarises the observational anchors, the model parameterisation, and the mechanism that produces a double-peaked lightcurve, and identifies the outstanding test: whether radiative heating of the metal-rich SN_A ejecta reproduces the observed absorption spectrum. All figures are reproducible from the pipeline data via `scripts/make_scenario_figures.py`.

1 Observational anchors

Figure 1 shows the ATLAS lightcurve of SN 2024pba in absolute magnitude (o and c bands). The two events are separated by ∼330–360 d: the first, fainter maximum (SN_A) peaks near MJD 60513 at $M \approx -17.2$, and the main event (SN_B) peaks at MJD 60850 with $o \approx -18.2$, preceded by a ∼30–40 d precursor bump near MJD 60810–60830. Table 1 lists the adopted anchors and distance/extinction. The interval between the events implies that any material connecting them has had ∼1 yr of homologous expansion, which sets the radial scale of the interaction region ($\sim 10^{16}$ cm for typical ejecta velocities of a few 10^3 km s^{-1}).

Figure 2 shows the two available optical spectra. The DBSP spectrum (MJD 60523.5, ∼10 d after the SN_A peak) belongs to the first event; the ALFOSC spectrum (MJD 60875.1, +25 d after the SN_B peak) samples the main event. The scientific requirement on the scenario is that the strong H, He, O, Si, Ca and Fe absorption seen in these spectra be physically attributed — ion, transition, formation radius and velocity — and in particular be testable against the hypothesis that the lines form in the radiatively heated SN_A ejecta rather than in the SN_B shell/CSM alone.

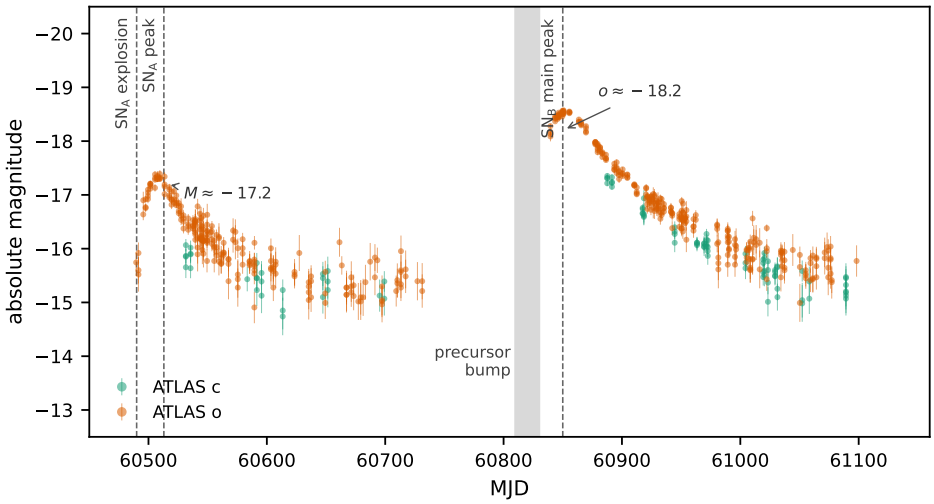


Figure 1: ATLAS *o*- (orange) and *c*-band (cyan) lightcurve of SN 2024pba in absolute magnitude (D_L = 80.4 Mpc, $E(B{-}V)$ = 0.15). Dashed lines mark the adopted anchors: SN_A explosion (∼MJD 60490, constrained by the ATLAS rise), SN_A peak (60513), and the SN_B main peak (60850). The grey band marks the precursor-bump window, ∼30–40 d before the main peak. A few bad-subtraction outliers have been removed.

2 The two events

Figure 3 illustrates the adopted geometry at the time of the second event.

2.1 SN_A — the first (Type IIb) explosion

SN_A is represented by the model `SN_A_g_T20k_Ni06`: a stripped KEPLER progenitor (`s12.0-presn-stripped`, mass cut 1.4 $M_\odot$) with a thin extended hydrogen envelope (0.05 $M_\odot$, R = 200 $R_\odot$, $T_{\rm base}$ = 20 kK), explosion energy ∼9 B and 0.06 $M_\odot$ of ^{56}Ni. The explosion produces fast, metal-rich (He/C/O/Si/Ca/Fe) ejecta that expand homologously for ∼1 yr before the second event. Figure 4 compares the STELLA (**??**) lightcurve of this model with the observed first peak: the model reproduces the observed ∼ −17 mag maximum scale.

2.2 SN_B — the second event

SN_B is represented not as a fresh full ejecta but as a *dense shell* ($M_{\rm shell} \approx 5$–$10\ M_\odot$, $v_{\rm shell} \approx 10^4$ km s^{-1}, solar composition) launched into the environment of the aging SN_A ejecta. The region between the shell and the SN_A ejecta inner edge is filled with a steady wind ($\dot{M}_{\rm wind} \approx 10^{-3}$–$3 \times 10^{-2}\ M_\odot\,{\rm yr}^{-1}$,

Table 1: Observational anchors and adopted distance/reduction.

Quantity	Value	Note
SN_A explosion	∼MJD 60490	ATLAS rise of first event
SN_A peak	MJD 60513, $M \approx -17.2$	fainter first maximum
SN_B precursor bump	∼MJD 60810–60830, $M \approx -15.8$	shock-cooling bump
SN_B main peak	MJD 60850, $o \approx -18.2$	lightcurve reference
D_L	80.4 Mpc (DM = 34.53)	$z = 0.0186$
$E(B-V)$	0.15	$A_o \approx 2.40\,E$
SN_A expl. → SN_B peak	≈ 360 d	
Interaction onset	∼320–340 d after SN_A	bump 30–40 d pre-peak

$v_{\rm wind} = 100$ km s^{-1}, $\rho \propto r^{-2}$) plus a smoothed wind→ejecta buffer controlled by `we_transition_dex`.

Figure 5 shows the actual initial structure fed to STELLA for the completed model `T6k_M10.0_V10000`. The four regions — solar-composition shell, $\rho \propto r^{-2}$ wind gap, the wind→ejecta buffer, and the metal-rich SN_A ejecta — are clearly separated in both density and composition. The velocity field is deliberately non-monotonic: the shell peaks at 10^4 km s^{-1}, the wind gap is nearly stationary, and the SN_A ejecta resumes homologous expansion ($v \propto r$) reaching 10^4 km s^{-1} at the outer edge.

3 What makes the lightcurve double-peaked

Figure 6 shows the STELLA lightcurve of `T6k_M10.0_V10000` against the observations, phased to each curve's own maximum. Two components are visible in both model and data:

1. **First bump** ($M \approx -16.3$, $t \approx 10$–40 d after launch): shock-cooling/diffusion. The shell launched at $\sim 10^4$ km s^{-1} immediately shocks the optically thick wind; the deposited energy diffuses out on the ∼30 d wind diffusion time. Its duration is set by the shock crossing time of the wind gap ($\sim 4 \times 10^{14} \rightarrow \sim 8 \times 10^{15}$ cm at 3000–5000 km s^{-1}). This corresponds to the observed precursor bump (~ -15.8, grey band).

2. **Main peak** ($M \approx -19.4$ at $t \approx 55$–60 d, $T_{\rm bb} \approx 10^4$ K): the embedded shock rams into the dense wind–ejecta buffer, where the density rises by ∼1 dex across the transition, boosting the dissipation ($\propto \rho v^3$) while the remaining wind column ahead of the shock goes optically thin and releases the stored radiation.

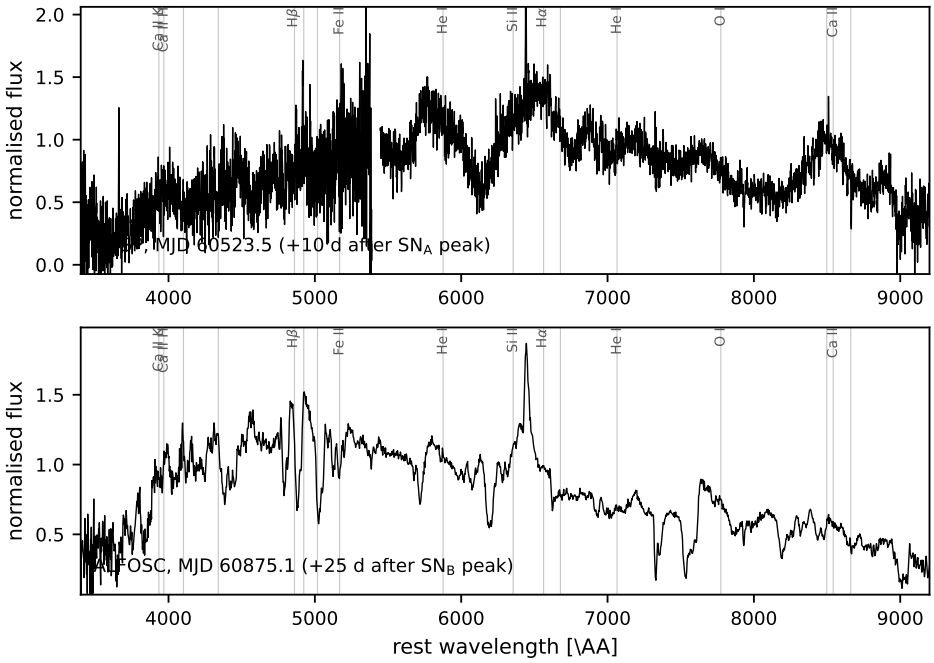


Figure 2: Observed spectra of SN 2024pba, rest-frame ($z = 0.0186$), normalised to the 5000–7000 Å median. Top: Palomar DBSP spectrum at MJD 60523.5, ten days after the SN_A peak. Bottom: NOT ALFOSC spectrum at MJD 60875.1, 25 d after the SN_B main peak. Grey lines mark the rest wavelengths of the key H, He, O, Si, Ca and Fe features.

3.1 Radiative heating, not shock heating, at maximum

The distinguishing feature of the scenario is that *at main peak the SN_A ejecta is radiatively heated, not shocked.* Figure 7 shows the radial structure of `T6k_M10.0_V10000` at $t = 20$, 40, 57 and 75 d from the completed STELLA run:

- The forward shock — visible as the velocity discontinuity (dotted lines) — sits at $r \approx 3$–7×10^{15} cm, roughly 10^{16} cm *inside* the metal-rich ejecta inner edge (dashed lines, $X_O > 0.05$) throughout this interval; in this model true shock–ejecta contact has not occurred by $t \approx 85$ d.
- The gas temperature at the ejecta face (dots, bottom panel) rises from ~ 2800 K at $t = 20$ d to ~ 6900 K at $t = 40$ d and remains ~ 5400 K at maximum ($t = 57$ d), well above the ~ 2200 K adiabatic floor of the unshocked material — the signature of the radiative precursor. Note $T_{gas} > T_{rad}$ throughout the gap and ejecta: the heating is deposited in the gas, and a nebular/dilute-LTE ionisation estimate would be inappropriate.

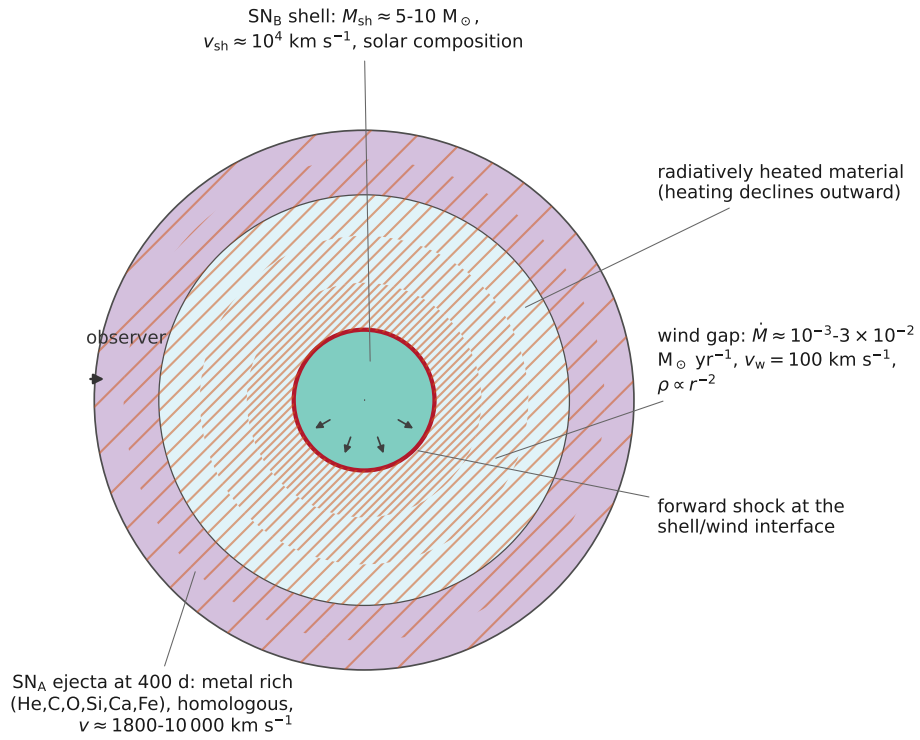


Figure 3: Schematic of the two-event geometry at SN_B shell launch (SN_A age $\approx$ 400 d; the wind→ejecta buffer is omitted for clarity). From inside out: the SN_B shell (solar composition, $M_{sh} \approx 5$–$10\ M_\odot$, $v_{sh} \approx 10^4$ km s^{-1}); the wind gap ($\rho \propto r^{-2}$); and the metal-rich, homologously expanding SN_A ejecta. The forward shock (red) sits at the shell/wind interface; the orange hatching marks the radiatively heated material ahead of it, with the heating declining with distance from the interface. The SN_A ejecta is heated radiatively, not shocked.

- The ejecta velocity field stays homologous ($v \propto r$ across the whole O-rich region); no propagating velocity discontinuity reaches the metal-rich gas during the photospheric phase.

The metal-rich ejecta therefore acts as a *radiatively heated screen* at unperturbed SN_A velocities ($\sim$ 1800–10 000 km s^{-1}) — the proposed source of the observed H, He, O, Si, Ca and Fe absorption. Whether the heating is sufficient to produce the observed line strengths at the observed velocities is the pending CMFGEN test (Section 5).

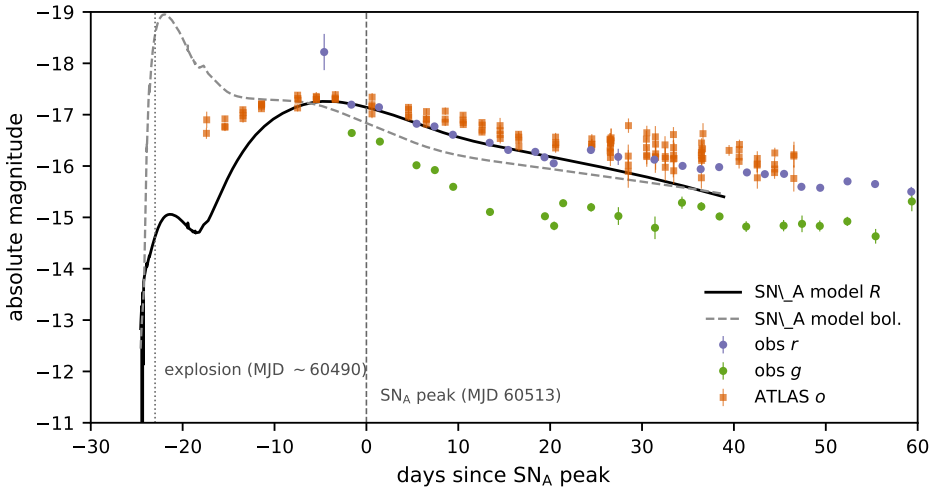


Figure 4: SN_A model `SN_A_g_T20k_Ni06` R-band (solid) and bolometric (dashed) lightcurve, with the explosion epoch aligned to MJD 60490, compared to the observed r, g and ATLAS o photometry around the first peak. The model reaches $M_R \approx -17.3$, close to the observed -17.2; the early bright phase is shock breakout through the extended envelope.

4 Model parameterisation: the `X_*` grid

All production models are generated and submitted through `scripts/run_model_grid.py`, with the naming convention `X_M{mass}_V{velocity}[_W{mdot}][_we{dex}]`. Table 2 lists the explored parameters; the SN_A ejecta age at shell launch is fixed at 400 d by the observed event separation.

Figure 8 compares the current grid to the observations. The parameter roles established so far are: peak height ← shell kinetic energy; rise to main peak ← `we_transition_dex` (sharper interface → faster turn-on of dissipation); post-peak decline ← wind column (higher $\dot{M}$ → longer diffusion and sustained ρv^3 dissipation); first-bump duration ← wind gap size, fixed by the 400 d SN_A age and modified by $\dot{M}$ and shell energetics.

5 Open points and caveats

- **Timeline convention.** `source_age_days=400` places the shell launch ∼40 d after the observed SN_B peak if taken literally against the MJD 60490 explosion date; the physically meaningful constraint is the ∼330–360 d gap between the two events. Model phases are always plotted relative to each model's own main peak.
- **Band mismatch.** Model curves are Johnson–Cousins R/V from

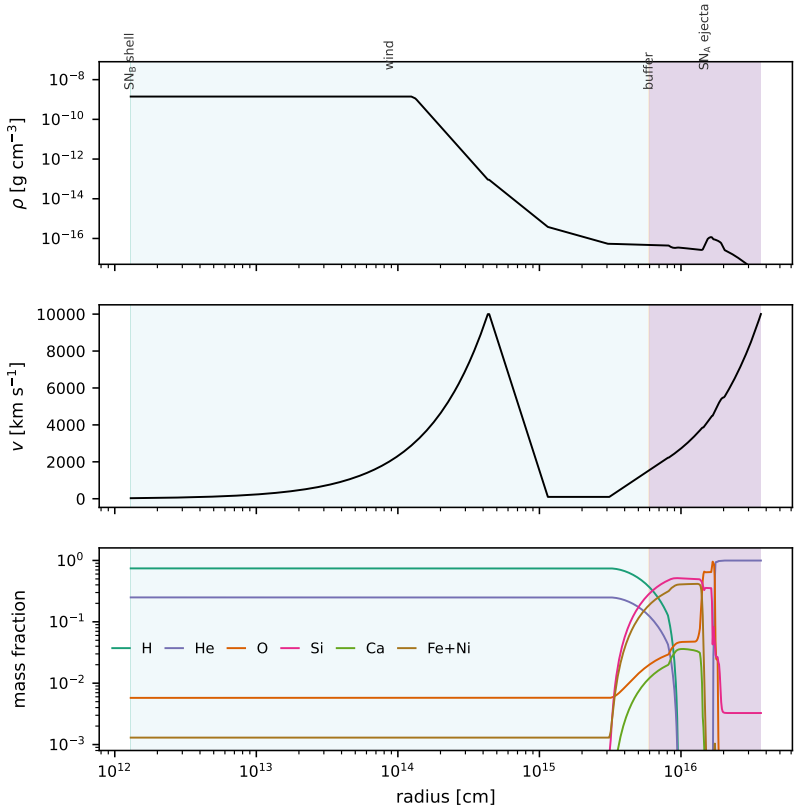


Figure 5: Initial STELLA input structure of `T6k_M10.0_V10000`: density (top), velocity (middle) and mass fractions (bottom) versus radius. Shading marks the SN_B shell (teal), the wind gap (blue), the wind→ejecta buffer (orange) and the metal-rich SN_A ejecta (purple). The ejecta carries the He/C/O/Si/Ca/Fe reservoir whose radiative heating is the key spectral hypothesis.

STELLA; the observations are SDSS r/g and ATLAS o. Expect 0.1–0.3 mag systematic offsets.

- **No ^{56}Ni tail.** SN_B ^{56}Ni is set to 10^{-4} $M_\odot$; if the observed tail is underluminous after the wind/grid exploration, a real ^{56}Ni floor ($\sim$ 0.02–0.05 $M_\odot$) is a separate lever.
- **Spectral attribution pending.** The scenario's key test — whether the radiatively heated SN_A ejecta produces the observed metal absorption at SN_A velocities — awaits CMFGEN (**??**) snapshots from the selected lightcurve model. Line identification (ion, transition, formation depth, velocity) and a comparison of SN_A-only, SN_B-only and interaction spectra are required before the hypothesis can be accepted.

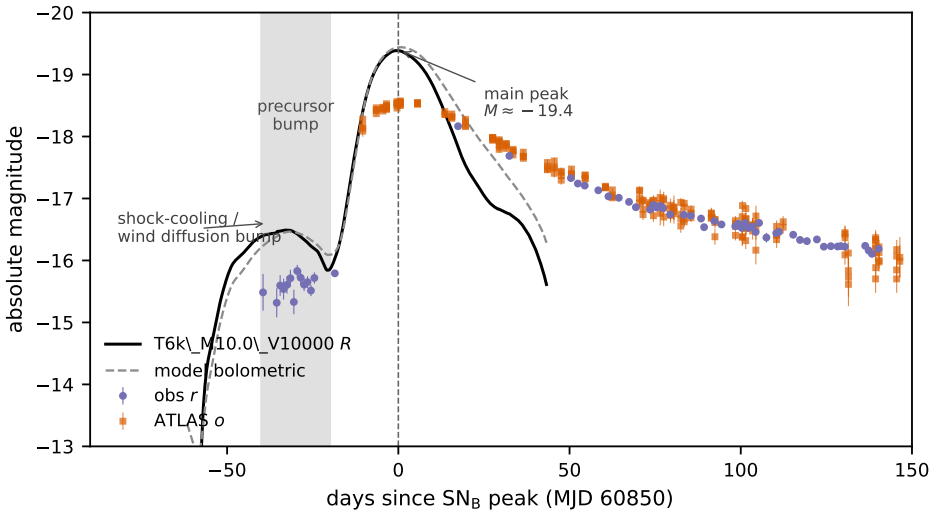


Figure 6: `T6k_M10.0_V10000` R-band (solid) and bolometric (dashed) lightcurve versus the observed r and ATLAS o photometry, phased to each curve's own maximum (model bands are Johnson–Cousins; systematic offsets of 0.1–0.3 mag vs. SDSS/ATLAS are expected). The model reproduces the two-component morphology: a ~ -16.3 mag cooling/diffusion bump followed by a ~ -19.4 mag interaction-powered main peak.

Table 2: Grid parameters and their established lightcurve roles.

Parameter	Values	Controls
`shell_mass_msun`	5, 7, 10	peak height (with $v_{\rm sh}$)
`shell_velocity_kms`	6000–10000	peak height
`wind_mdot_msun_yr`	10^{-3}, 10^{-2}, 3×10^{-2}	post-peak decline
`we_transition_dex`	0.5, 0.15	rise to main peak
`source_age_days`	400 (fixed)	wind gap size / bump duration
^{56}Ni in shell	10^{-4} $M_\odot$	negligible by construction

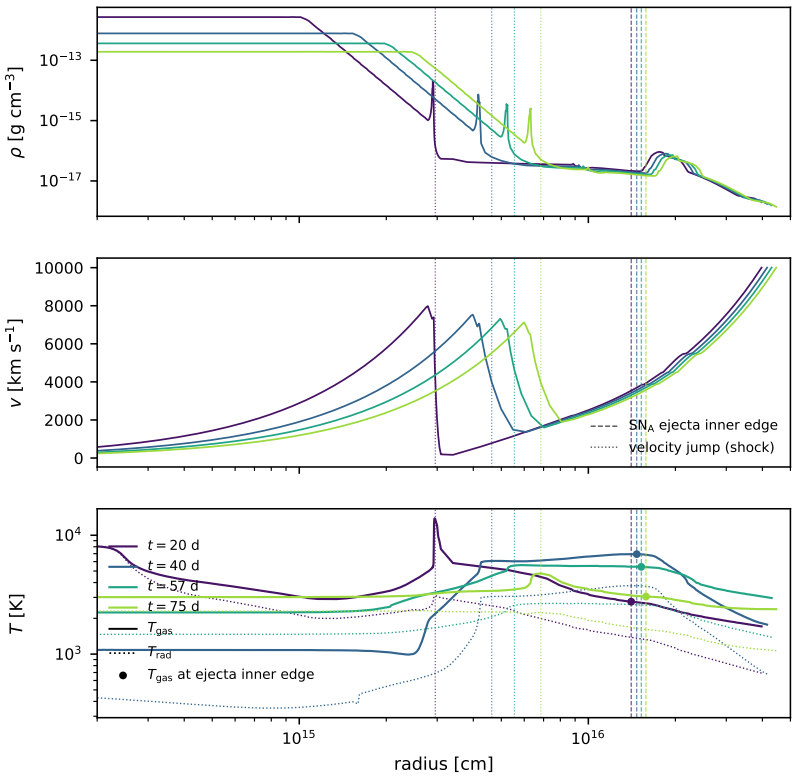


Figure 7: Radial structure of `T6k_M10.0_V10000` from the STELLA `.swd` at $t = 20$, 40, 57 and 75 d after shell launch (colour-coded). Top: density. Middle: velocity. Bottom: gas temperature (solid) and radiation temperature (dotted). Dashed vertical lines mark the inner edge of the O-rich SN_A ejecta ($X_O > 0.05$); dotted lines mark the largest velocity discontinuity inside it (the embedded shock). The ejecta face is heated radiatively while its velocity field remains homologous.

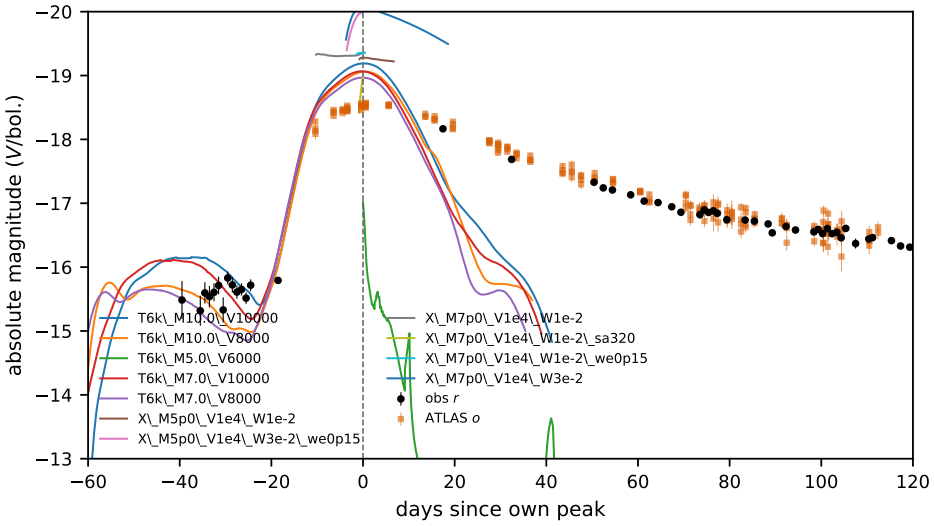


Figure 8: Grid lightcurves (model V or bolometric magnitude, phased to each model's own peak) versus the observed r-band and ATLAS o photometry. Shell kinetic energy sets the peak height, the wind column the post-peak decline, and `we_transition_dex` the sharpness of the rise. Some `X_*` runs are still in progress; truncated curves reflect completed STELLA time only.